



ICT711 PROGRAMMING AND ALGORITHMS T126

All information in the Subject Outline is correct at the time of approval. KOI reserves the right to make changes to the Subject Outline if they become necessary. Any changes require the approval of the KOI Academic Board and will be formally advised to those students who may be affected by email and via Moodle.

Information contained within this Subject Outline applies to students enrolled in the trimester as indicated

1. General Information

1.1 Administrative Details

Associated HE Award(s)	Duration	Level	Subject Coordinator
Master of Information Technology (MIT) Graduate Diploma of Information Technology (GDIT) Graduate Certificate of Information Technology (GCIT)	1 trimester	Postgraduate	Dr Saad Sajid Hashmi saad.hashmi@koi.edu.au P: +61 (2) 9283 3583 L: 7-11, 11 York Street. Consultation: via Moodle or by appointment.

1.2 Core/Elective

This subject is:

- a core subject for the Master of Information Technology (MIT)
- a core subject for the Graduate Diploma of Information Technology (GDIT)
- a core subject for the Graduate Certificate of Information Technology (GCIT)

1.3 Subject Weighting

Indicated below is the weighting of this subject and the total course points.

Subject Credit Points	Total Course Credit Points
4	MIT (64 Credit Points); GDIT (32 Credit Points); GCIT (16 Credit Points);

1.4 Student Workload

Indicated below is the expected student workload per week for this subject

No. Timetabled Hours/Week*	No. Personal Study Hours/Week**	Total Workload Hours/Week***
3 hours/week plus supplementary online material	7 hours/week	10 hours/week

* Total time spent per week at lectures and tutorials

** Total time students are expected to spend per week in studying, completing assignments, etc.

*** Combination of timetable hours and personal study

1.5 Mode of Delivery Classes will be face-to-face or hybrid. Certain classes will be online (e.g., special arrangements).

1.6 Pre-requisites Nil



1.7 General Study and Resource Requirements

- Students are expected to attend classes with the weekly worksheets and subject support material provided in Moodle. Students should read this material before coming to class to improve their ability to participate in the weekly activities.
- Students will require access to the internet and their KOI email and should have basic skills in word processing software such as MS Word, spreadsheet software such as MS Excel and visual presentation software such as MS PowerPoint.
- Computers and WIFI facilities are extensively available for student use throughout KOI. Students are encouraged to make use of the campus Library for reference materials.

Software resource requirements specific to this subject: JDK 8 and NetBeans IDE, Eclipse IDE, Office 365, MS Imagine.

1.8 Academic Advising

Academic advising is available to students throughout teaching periods including the exam weeks. As well as requesting help during scheduled class times, students have the following options:

- Consultation times: A list of consultation hours is provided on the homepage of Moodle where appointments can be booked.
- Subject coordinator: Subject coordinators are available for contact via email. The email address of the subject coordinator is provided at the top of this subject outline.
- Academic staff: Lecturers and Tutors provide their contact details in Moodle for the specific subject. In most cases, this will be via email. Some subjects may also provide a discussion forum where questions can be raised.
- Head of Program: The Head of Program is available to all students in the program if they need advice about their studies and KOI procedures.
- Vice President (Academic): The Vice President (Academic) will assist students to resolve complex issues (but may refer students to the relevant lecturers for detailed academic advice).

2. Academic Details

2.1 Overview of the Subject

This subject extends the coverage of programming basics to concepts of exception handling, advanced arrays, inheritance for program design and implementation in Java. Program design and complexities of different algorithms are evaluated to assist in problem solving. An object-oriented framework is employed to develop a deeper knowledge of computational problem-solving through the use of established algorithms and programming methodologies, strategies and techniques.

2.2 Graduate Attributes for Postgraduate Courses

Graduates of postgraduate courses from King's Own Institute will achieve the graduate attributes expected from successful completion of a postgraduate degree under the Australian Qualifications Framework (2nd edition, January 2013). Graduates at this level will be able to apply advanced body of knowledge from their major area of study in a range of contexts for professional practice or scholarship and as a pathway for further learning.

King's Own Institute's generic graduate attributes for a master's level degree are summarised below:

	KOI Postgraduate Degree Graduate Attributes	Detailed Description
	Knowledge	Current, comprehensive and coherent knowledge, including recent developments and applied research methods
	Critical Thinking	Critical thinking skills to identify and analyse current theories and developments and emerging trends in professional practice
	Communication	Communication and technical skills to analyse and theorise, contribute to professional practice or scholarship, and present ideas to a variety of audiences
	Research and Information Literacy	Cognitive and technical skills to access and evaluate information resources, justify research approaches and interpret theoretical propositions
	Creative Problem Solving Skills	Cognitive, technical and creative skills to investigate, analyse and synthesise complex information, concepts and theories, solve complex problems and apply established theories to situations in professional practice
	Ethical and Cultural Sensitivity	Appreciation and accountability for ethical principles, cultural sensitivity and social responsibility, both personally and professionally
	Leadership and Strategy	Initiative, leadership skills and ability to work professionally and collaboratively to achieve team objectives across a range of team roles Expertise in strategic thinking, developing and implementing business plans and decision making under uncertainty
	Professional Skills	High level personal autonomy, judgement, decision-making and accountability required to begin professional practice

Across the course, these skills are developed progressively at three levels:

- **Level 1 Foundation** – Students learn the skills, theories and techniques of the subject and apply them in stand-alone contexts
- **Level 2 Intermediate** – Students further develop skills, theories and techniques of the subject and apply them in more complex contexts, beginning to integrate the application with other subjects.
- **Level 3 Advanced** – Students have a demonstrated ability to plan, research and apply the skills, theories and techniques of the subject in complex situations, integrating the subject content with a range of other subject disciplines within the context of the course.

Generally, skills gained from subjects in the Graduate Certificate and Graduate Diploma are at levels 1 and 2 while other subjects in the Master's degree are at level 3.

2.3 Subject Learning Outcomes

Listed below, are key knowledge and skills students are expected to attain by successfully completing this subject:

Subject Learning Outcomes	Contribution to Graduate Attributes
a) Apply object-oriented programming principles, methods, and techniques using Java	   
b) Construct and evaluate different algorithms and data structures including searching and sorting algorithms and linked list structures	   
c) Critically analyse and design a medium-sized application according to the given requirements and communicate programming solutions and their derivation	      
d) Implement and test a medium-sized application according to the given requirements	     

2.4 Subject Content and Structure

Below are details of the subject content and how it is structured, including specific topics covered in lectures and tutorials. Reading refers to the text unless otherwise indicated.

Weekly Planner:

Week (beginning)	Topic covered in each week's lecture	Reading(s)	Expected work as listed in Moodle
Week 1 02 March	Fundamental of Java Programming (Data types, Conditions, Loops, Methods)	Chs. 1, 2, 3, 5, 6 [J. Farrell 2023]	Discussion on Java application individual and group project Tutorial exercises on designing and running Java programs (Data types, operators, control structures and methods) Formative not graded
Week 2 09 March	Object-Oriented Programming (Classes, Objects)	Ch. 4 [J. Farrell 2023] Ch 2 [D S. Myers 2025]	Discussion on Java application group project Tutorial exercises on creating and manipulating simple classes and objects Summative graded
Week 3 16 March	Introduction to inheritance	Ch. 9 [J. Farrell 2023]	Discussion on Java application group project Tutorial exercises on diagramming inheritance using UML.



Week (beginning)	Topic covered in each week's lecture	Reading(s)	Expected work as listed in Moodle
			Summative graded
Week 4 23 March	Advanced inheritance concepts	Ch. 9 [J. Farrell 2023]	Discussion on Java application group project Tutorial exercises on designing class hierarchies using abstract classes and interfaces Summative graded
Week 5 30 March	Exception and File IO	Ch. 10 and 11 [J. Farrell 2023]	Discussion on Java application group project Tutorial exercises on exception, file IO and testing code Summative graded Assessment 2: Quiz due
Week 6 07 April (Tue)	Data Structures Part I (Arrays, ArrayLists, Stack, Queue)	Ch. 8 [J. Farrell 2023] Ch. 2, [J. Cutajar 2018]	Discussion on Java application group project Tutorial exercises on building and manipulating arrays, ArrayLists, stacks, and queues Summative graded
Week 7 13 April	Data Structures Part II (LinkedList, Binary Trees)	Ch. 2 and 3 [J. Cutajar 2018] Ch 6 and 17 [D S. Myers 2025]	Discussion on Java application group project Tutorial exercises on constructing linked lists and implementing binary trees Summative graded
Week 8 20 April	Graphical User Interface (GUI)	Ch. 14 [J. Farrell 2023]	Discussion on Java application individual project Tutorial exercises on GUI Summative graded Assessment 3 due
Week 9 28 April (Tue)	Algorithm Complexity and Efficiency (Big-O Notation, Time Complexity, Space Complexity)	Ch. 1 [J. Cutajar 2018]	Discussion on Java application group project Tutorial exercises on measuring algorithmic complexity and identifying different complexities Summative graded



Week (beginning)	Topic covered in each week's lecture	Reading(s)	Expected work as listed in Moodle
Week 10 04 May	Sorting Algorithms - Basic Sorting and Efficient Sorting (Bubble Sort, Merge Sort, Quick Sort)	Ch. 2 [J. Cutajar 2018] Ch 10 [D S. Myers 2025]	Discussion on Java application individual project Tutorial exercises on implementing and comparing sorting algorithms; discussion on performance trade-offs Summative graded
Week 11 11 May	Searching Algorithms and Hash Tables (Linear Search, Binary Search, Hash Tables)	Ch. 3 [J. Cutajar 2018] Ch 5, 15 and 17 [D S. Myers 2025]	Discussion on Java application individual project Tutorial exercises on implement search algorithms and hash table operations Summative graded Assessment 4 due
Week 12 18 May	Testing Technique	Recommended readings	Tutorial exercises on writing test cases and debugging code Assessment 4: Project demonstration due
Week 13 25 May	Study review week and Final Exam Week		
Week 14 01 June	Examinations Continuing students - enrolments for T226 open		Please see exam timetable for exam date, time and location
Week 15 09 June (Tue)	Student Vacation begins New students - enrolments for T226 open		
Week 16 15 June	- Results Released - Review of Grade Day for T126 – see Sections 2.6 and 3.2 below for relevant information. - Certification of Grades NOTE: More information about the dates will be provided at a later date through Moodle/KOI email.		
T226 29 June 2026			
Week 1 29 June	Week 1 of classes for T226		

2.5 Public Holiday Amendments

Please note: KOI is closed on all scheduled NSW Public Holidays.



T126 has Five (5) public holidays that occur during this trimester. Classes scheduled for these public holidays (Calendar Class Date) will be rescheduled as per the table below.

This applies to ALL subjects taught in T126.

Please see the table below and adjust your class timing as required. Please make sure you have arrangements in place to attend the rescheduled classes if applicable to your T126 enrolment.

Classes will be conducted at the same time and in the same location as your normally scheduled class except these classes will be held on the date shown below.

Calendar Class Date	Rescheduled Class Date
03 April 2026 (Friday) 04 April 2026 (Saturday) 06 April 2026 (Monday) 25 April 2026 (Saturday) 27 April 2026 (Monday)	25 May 2026 (Monday) 26 May 2026 (Tuesday) 27 May 2026 (Wednesday)

2.6 Review of Grade, Deferred Exams & Supplementary Exams/Assessments

Review of Grade:

There may be instances when you believe that your final grade in a subject does not accurately reflect your performance against the marking criteria. Section 8 of the *Assessment and Assessment Appeals Policy* (www.koi.edu.au) describes the grounds on which you may apply for a Review of Grade.

If you have a concern about your marks and you are unable to resolve it with the Academic staff concerned, then you can apply for a formal Review of Grade as explained in section 3.2(e) Appeals Process below. Please note the time limits for requesting a review. Please ensure you read the Review of Grade information before applying.

Review of Grade Day:

Final exam scripts will not normally be returned to students. Students can obtain feedback on their exam performance and their results for the whole subject at the Review of Grade Day. KOI will hold the Review of Grade Day for all subjects studied in T126. **The ROG day will be announced at a later date and the students will be notified through Moodle/KOI email.**

Only final exams and whole subject results will be discussed as all other assessments should have been reviewed during the trimester. Further information about Review of Grade Day will be available through Moodle.

If you fail one or more subjects and you wish to consider applying for a Review of Grade you are **STRONGLY ADVISED to** attend the Review of Grade Day. You will have the chance to discuss your final exam and subject result with your lecturer, and will be advised if you have valid reasons for applying for a Review of Grade (see Section 3.2 below and the *Assessment and Assessment Appeals Policy*).

A formal request for a review of grade may not be considered unless you first contact the subject coordinator to discuss the result.

Deferred Exams:

If you wish to apply for a deferred exam because you are unable to attend the scheduled exam, you should submit the Assignment Extension / Exam Deferment Form available by contacting



academic@koi.edu.au as soon as possible, *but no later than three (3) working days of the assessment due date.*

If you miss your mid-trimester or final exam there is no guarantee you will be offered a deferred exam.

You must apply within the stated timeframe and satisfy the conditions for approval to be offered a deferred exam (see Section 8.1 of the *Assessment and Assessment Appeals Policy and the Application for Assignment Extension or Deferred Exam Forms*). In assessing your request for a deferred exam, KOI will take into account the information you provide, the severity of the event or circumstance, your performance on other items of assessment in the subject, class attendance and your history of previous applications for special consideration.

Deferred mid-trimester exams will be held before the end of week 9. Deferred final exams will be held on two days during week 1 or 2 in the next trimester. You will not normally be granted a deferred exam on the grounds that you mistook the time, date or place of an examination, or that you have made arrangements to be elsewhere at that time; for example, have booked plane tickets.

If you are offered a deferred exam, but do not attend *you will be awarded 0 marks for the exam*. This may mean it becomes difficult for you to pass the subject. If you apply for a deferred exam within the required time frame and satisfy the conditions you will be advised by email (to your KOI student email address) of the time and date for the deferred exam. Please ensure that you are available to take the exam at this time.

Marks awarded for the deferred exam will be the marks awarded for that item of assessment towards your final mark in the subject.

Supplementary Assessments (Exams and Assessments):

A supplementary assessment may be offered to students to provide a final opportunity to demonstrate successful achievement of the learning outcomes of a subject. *Supplementary assessments are only offered at the discretion of the Board of Examiners*. In considering whether or not to offer a supplementary assessment, KOI will take into account your performance on all the major assessment items in the subject, your attendance, participation and your history of any previous special considerations.

If you are offered a supplementary assessment, you will be advised by email to your *KOI student email address* of the time and due date for the supplementary assessment – supplementary exams will normally be held at the same time as deferred final exams during week 1 or week 2 of the next trimester.

You must pass the supplementary assessment to pass the subject. The maximum grade you can achieve in a subject based on a supplementary assessment is a PASS grade.

If you:

- are offered a supplementary assessment, but fail it;
 - are offered a supplementary exam, but do not attend; or
 - are offered a supplementary assessment but do not submit by the due date;
- you will receive a FAIL grade for the subject.

Students are also eligible for a supplementary assessment for their final subject in a course where they fail the subject but have successfully completed all other subjects in the course. You must have completed all major assessment tasks for the subject and obtained a passing mark on at least one of the major assessment tasks to be eligible for a supplementary assessment.

If you believe you meet the criteria for a supplementary assessment for the final subject in your course, but have not received an offer, complete the *Complaint, Grievance, Appeal Form* and send your form to reception@koi.edu.au. The deadline for applying for supplementary assessment is the Friday of the first week of classes in the next trimester.



2.7 Teaching Methods/Strategies

Briefly described below are the teaching methods/strategies used in this subject:

- o *Lectures* (1 hours/week) are conducted in seminar style and address the subject content, provide motivation and context and draw on the students' experience and preparatory reading.
- o *Tutorials* (2 hours/week) include class discussion of case studies and research papers, practice sets and problem-solving and syndicate work on group projects. Tutorials often include group exercises and so contribute to the development of teamwork skills and cultural understanding. Tutorial participation is an essential component of the subject and contributes to the development of many of the graduate attributes (see section 2.2 above). Tutorial participation contributes towards the assessment in many subjects (see details in Section 3.1 for this subject). Supplementary tutorial material such as case studies, recommended readings, review questions etc. will be made available each week in Moodle.
- o *Online* teaching resources include class materials, readings, model answers to assignments and exercises and discussion boards. All online materials for this subject as provided by KOI will be found in the Moodle page for this subject. Students should access Moodle regularly as material may be updated at any time during the trimester
- o *Other contact* - academic staff may also contact students either via Moodle messaging, or via email to the email address provided to KOI on enrolment.

2.8 Student Assessment

Assessment is designed to encourage effective student learning and enable students to develop and demonstrate the skills and knowledge identified in the subject learning outcomes. Assessment tasks during the first half of the study period are usually intended to maximise the developmental function of assessment (formative assessment). These assessment tasks include weekly tutorial exercises (as indicated in the weekly planner) and low stakes graded assessments (as shown in the graded assessment table). The major assessment tasks where students demonstrate their knowledge and skills (summative assessment) generally occur later in the study period. These are the major graded assessment items shown in the graded assessment table.

Final grades are awarded by the Board of Examiners in accordance with KOI's Assessment and Assessment Appeals Policy. The definitions and guidelines for the awarding of final grades are:

- o *HD High distinction* (85-100%): an outstanding level of achievement in relation to the assessment process.
- o *D Distinction* (75-84%): a high level of achievement in relation to the assessment process.
- o *C Credit* (65-74%): a better than satisfactory level of achievement in relation to the assessment process.
- o *P Pass* (50-64%): a satisfactory level of achievement in relation to the assessment process.
- o *F Fail* (0-49%): an unsatisfactory level of achievement in relation to the assessment process.
- o *FW*: This grade will be assigned when a student did not submit any of the compulsory assessment items.

Provided below is a schedule of formal assessment tasks and major examinations for the subject.

Assessment Type	When Assessed	Weighting	Learning Outcomes Assessed
Assessment 1: Tutorial exercises	Weeks 2-11	20%	a, b
Assessment 2: Quiz	Week 5	10%	a



Assessment Type	When Assessed	Weighting	Learning Outcomes Assessed
Assessment 3: Java application Group Project (1500-2000 words)	Week 8	Group work: 25% Individual contribution: 10% Total: 35%	a, c, d
Assessment 4: Individual Assignment (2000-2500 words)	Week 11- Week 12	35%	a, b, c, d

Requirements to Pass the Subject:

To gain a pass or better in this subject, students must gain a *minimum of 50%* of the total available subject marks.

2.9 Prescribed and Recommended Readings

Provided below, in formal reference format, is a list of the prescribed and recommended readings.

Prescribed Texts:

Farrell, J (2023), *Java Programming*, 10th ed. Cengage, Mason OH. Available at : ProQuest Ebook. ISBN: 9780357673553.

Myers, D. S (2025), *Data Structures and Algorithms in Java: A Project-Based Approach*, Cambridge University Press. ISBN: 9781009260343.

Recommended Readings:

Cutajar, J (2018), *Beginning Java Data Structures and Algorithms: Sharpen Your Problem-Solving Skills by Learning Core Computer Science Concepts in a Pain-Free Manner*, Packt Publishing, Limited, Birmingham. Available at : ProQuest Ebook Central. ISBN: 9781789533293.

Langr, J., (2024). *Pragmatic Unit Testing in Java with JUnit*. Pragmatic Bookshelf. ISBN: 9798888651032.

Campesato, O (2023), *Data Structures in Java*, Mercury Learning & Information, Bloomfield. Available at : ProQuest Ebook Central. ISBN: 9781683929420.

Evans, B. J., Clark, J., & Flanagan, D. (2023). *Java in a Nutshell*. O'Reilly Media, Inc.. ISBN: 9781098130992.

Bouras, A. (2024) *Java and Algorithmic Thinking for the Complete Beginner*. Birmingham: Packt Publishing. ISBN: 9781835084939.

Suggested Periodicals:

- o The Programming Journal: <http://programming-journal.org/>
- o Science of Computer Programming: <https://www.journals.elsevier.com/science-of-computer-programming>
- o International Journal of Programming Languages and Applications: <http://airccse.org/journal/ijpla/ijpla.html>

Useful Websites:

- o <https://www.tutorialspoint.com/java/>
- o <https://www.w3schools.com/java/>
- o <https://www.codecademy.com/learn/learn-java>
- o <https://www.learnjavaonline.org/>

Conference/ Journal Articles:

Students are encouraged to read peer reviewed journal articles and conference papers. Google Scholar provides a simple way to broadly search for scholarly literature. From one place, you can search across many disciplines and sources: articles, theses, books, abstracts and court opinions, from academic publishers, professional societies, online repositories, universities and other web sites.

3. Assessment Details

3.1 Details of Each Assessment Item

The assessments for this subject are described below. The description includes the type of assessment, its purpose, weighting, due date and submission requirements, the topic of the assessment, details of the task and detailed marking criteria, including a marking rubric for essays, reports and presentations. Supplementary assessment information and assistance can be found in Moodle.

KOI expects students to submit their own work in both assignments and exams, or the work of their group in the case of group assignments. Material in assignments which comes from the work of others must be appropriately acknowledged.

Assessment 1

Assessment Type: Weekly tutorial exercises - individual assessment

Purpose: Weekly tutorial exercises are designed to encourage engagement and to develop and reinforce the knowledge and skills presented in the lectures. This assessment contributes to learning outcomes a and b.

Assessment topic: Java exercises

Task details: Weekly tutorial exercises from weeks 2-11 are submitted on Moodle every week (the week after the tutorial). The questions vary according to the weekly topic.

Value: 20% (2% per week) Due date: Weeks 2-11

Submission requirements details: Submitted the completed tutorial on Moodle.

Grading Guide Assessment 1: 20%

Criteria	Unsatisfactory	Satisfactory	Effective	Excellent	Exceptional
Weekly score out of 20 (Each week 2 marks)	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)

Assessment 2

Assessment type: Quiz (In class, face-to-face)



Purpose: This assessment will allow students to demonstrate their understanding of the concepts covered in weeks 1 to 4. This assessment contributes to learning outcome a.

Value: 10%

Due date: Week 5

Assessment topic: Java exercises

Task details: The quiz will consist of questions and problems relating to subject content from weeks 1 – 4 inclusive.

Submission requirements details: The quiz needs to be completed on Moodle during the in-class, face-to-face session.

Grading Guide Assessment 2: 10%

Criteria	Unsatisfactory	Satisfactory	Effective	Excellent	Exceptional
Total score out of 10	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)

Assessment 3

Assessment type: Java application – Group Project

This assessment will allow students to demonstrate their understanding of the concepts covered in Weeks 1 to 7. This assessment contributes to learning outcomes a, c and d.

In this group project, students will collaboratively design, implement, and test a text-based application using Java. The project requires applying object-oriented programming principles such as inheritance, polymorphism, abstraction, and encapsulation to develop a structured and maintainable application.

On successful completion of this project, you should have demonstrated that you are able to:

- Write a program consisting of multiple interacting objects with inheritance and use of polymorphism and abstraction
- Write a program that provides the requested functionality for the system
- Design class hierarchies to promote code reusability and extend base classes for specific functionality
- Utilise appropriate collection classes from Java's Collections API in appropriate ways/places
- Utilise advanced data structures like ArrayList, Linked Lists, Queues, etc. for efficient data manipulation and management
- Read data from and write data to text files
- Handle exceptions by constructing try ... catch blocks for appropriate circumstances
- Design a text-based user interface

Value: 35% (Group work 25%; Individual contribution 10%)

Due Date: Week 8

Assessment topic: The general idea is to read data about specific objects from a text file, store them in a Java collection, execute a set of CRUD (Create, Read, Update, Delete) instructions based on user input, and save the updated data back to a text file. Advanced data structures should be used to handle dynamic data efficiently and optimise system performance.

In addition to the group-based project, each member will be required to participate in a peer review process. After submission of the final group project, each student will evaluate the contributions of their peers. Individual grades may be adjusted based on the results of the peer review.

Task details: Provided on Moodle



Submission requirements details: You need to upload the following documents on Moodle

- Java source code files
- A project report detailing the system design, implementation, and testing
- A weekly activity log documenting contributions from each group member and peer review feedback.

Grading Guide Assessment 3: Group Work: 25%

Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Object-Oriented Design (2 marks)	Poor or incorrect use of OOP principles; fundamentally flawed design.	Basic class structure with major inaccuracies or lack of reusability.	Good design with one major inaccuracy or multiple minor issues.	Well-designed with minor logical or structural issues. Partially demonstrated in the report.	Strong object-oriented design with well-structured and reusable classes. Clearly demonstrated in the report.
Code Structure and Readability (1 mark)	Disorganized and difficult to read; poor indentation and no comments.	Somewhat structured but lacks clarity or comments.	Reasonably structured; minor readability issues.	Well-structured and readable; mostly clear with minor formatting issues. Partially demonstrated in the report.	Exceptionally well-organized, readable, and properly documented. Clearly demonstrated in the report using appropriate code snippets.
Add Record (3 marks)	Records are not correctly added or updated.	Basic functionality present but with major logic errors or missing validations.	Mostly correct, but several functional or logical issues exist.	Works correctly with minor issues, such as edge cases not handled. Partially demonstrated in the report.	Fully functional, correctly adds/updates records with validation and error handling. Clearly demonstrated in the report using file handling.
Delete Record (3 marks)	Records not correctly deleted	Acceptable code with some major issues with deletion	Deleted with several issues	Mostly correct with minor issues. Partially demonstrated in the report.	Records correctly deleted. Clearly demonstrated in the report using file handling.
Query Records (2 marks)	Queries do not work as expected	Some correct results; several issues	Mostly correct results; some minor issues	Return correct results; minor issues. Partially demonstrated in the report.	Queries return correct results for all criteria. Clearly demonstrated in report using file handling..
Reading/Writing Files (1 mark)	File operations do not work or are missing.	Basic file handling present but with several issues.	File operations mostly work, but have more than one minor issue.	File handling is correct, with minor issues in error handling. Partially demonstrated in the report.	Files are read and written correctly with appropriate error handling. Clearly demonstrated in report..



Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Saving Query Results (1 mark)	Query results not saved correctly	Saved with significant formatting issues	Mostly correct with minor formatting issues	Correct with minor formatting issues. Partially demonstrated in the report.	Query results saved correctly; proper formatting. Clearly demonstrated in the report using file handling.
Exception Handling (1 marks)	No exception handling or incorrect usage.	Basic try-catch usage, but with major issues or incorrect placements.	Good use of exception handling, but with some incorrect cases.	Mostly correct use, with minor improvements needed. Partially demonstrated in the report.	Effective and appropriate use of try-catch blocks with proper handling. Clearly demonstrated in the report.
Error Messages (1 mark)	Not informative or missing	Basic or somewhat unclear	Reasonable message that can be further improve in clarity	Mostly informative with minor issues. Partially demonstrated in the report.	Informative and user-friendly. Clearly demonstrated in the report.
Text-Based Interface (2 marks)	Confusing or non-functional interface	Basic interface; some usability issues	Good interface; few usability issues	Mostly clear; minor issues. Partially demonstrated in the report.	Clear and user-friendly; effective commands/prompts . Clearly demonstrated in the report.
Test Cases (3 marks)	Test cases are not provided; Inadequate or missing test cases	Basic test cases; several gaps	Good test cases; with few major gaps	Adequate with minor gaps. Partially demonstrated in the report.	Comprehensive and effective test cases. Clearly demonstrated in the report.
Screenshots and Report (5 marks)	Poor or missing screenshots and documentation	Basic; significant issues	Good with reasonable screenshots and documentation can be further improved. Shows a reasonable number of appropriate code snippets and outputs but missing figure captions and figure numbers.	Adequate with minor issues. Shows a reasonable number of appropriate code snippets and outputs with figure captions and figure numbers.	Well-organized; Shows most appropriate code snippets and outputs with figure captions and figure numbers.
Total mark out of 25					



Grading Guide Assessment 3: Individual Contribution: 10%

Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
Attendance at weekly meetings 2 marks	Attended less than 50% of the meetings	Attended less than 65% of the meetings	Attended less than 75% of the meetings	Attended less than 85% of the meetings	Attended 85% or more of the meetings
Weekly activity log 2 marks	Provided less than 50% of the weekly logs and/or provided scant details	Provided less than 65% of the weekly logs and/or provided incomplete details	Provided less than 75% of the weekly logs and/or provided only basic details	Provided less than 75% of the weekly logs and/or provided most required details	Provided less than 75% of the weekly logs and/or provided comprehensive details
Time management 2 marks	Assigned tasks typically late or not completed	Most assigned tasks completed by the deadline	All assigned tasks completed by the deadline	Most assigned tasks completed well in advance of deadline allowing time for revision and refinement	All assigned tasks completed well in advance of deadline allowing time for revision and refinement
Tasks 2 marks	Did not complete the tasks assigned	Completed most assigned tasks but not thorough and lacking details	Completed most assigned tasks in detail	Completed all assigned tasks in detail	Completed all tasks assigned at a professional level
Actual contribution to group project 2 marks	Made little or no contribution to the project/report	Made a cursory contribution to the project/report	Made a fair contribution to the project/report	Made a significant contribution to the project/report	Made a vital contribution to the project/report
Total mark out of 10					



Assessment 4

Assessment Type: Java application – Individual Project

Assessment topic: Algorithms and GUI

In this individual project, you will enhance the application developed in Assessment 3 by implementing both a Graphical User Interface (GUI) and a Text-Based Interface (TBI), allowing users to choose their preferred mode of interaction. Additionally, the application must incorporate sorting and searching algorithms to manage and retrieve data efficiently. You are required to apply appropriate testing techniques to validate the functionality, efficiency, and performance of the application. A detailed report must be submitted, outlining your design choices, testing approach, encountered challenges, and solutions implemented.

This assessment contributes to learning outcomes a, b, c, and d.

Value: 35%

Due Date: Week 11 – Week 12

Task details: Detailed task instructions will be provided on Moodle

Submission requirements details: You need to submit your work on Moodle and it should include

- **Code:** Well-structured Java program with appropriate comments and documentation.
- **Report:**
 - Project Overview (System description, objectives, key features).
 - User Interface Implementation (Details on GUI and text-based interface design).
 - System Architecture (Class diagrams, design patterns used).
 - Sorting & Searching Implementation (Algorithms applied and their justification).
 - Discussion on algorithm used and their complexities
 - Testing & Debugging (Test cases, expected vs. actual results, debugging logs).
 - Reflection (Challenges faced, problem-solving approach, and lessons learned).

During the Week 12 class, each student will be required to demonstrate their enhanced application to the class.

Grading Guide Assessment 4: Individual Project

Value: 35%

Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
GUI Design (2 marks)	Interface is poorly designed or non-functional	Basic GUI design; significant usability or layout issues	Functional GUI with minor usability or layout issues	Well-designed GUI with mostly clear layout and usability	Excellent, user-friendly, and well-structured GUI with no usability issues. Clearly demonstrated in the report..
GUI Functionality (3 marks)	GUI functions do not work as intended or fail to interact with the backend	GUI is functional but has significant issues in interaction	GUI is functional with minor interaction issues	GUI is mostly functional with smooth backend interaction; minor issues	Fully functional GUI with seamless backend interaction. Clearly demonstrated in the report..
Text-Based Interface (TBI) (2 marks)	TBI is missing or non-functional.	Basic TBI implemented but difficult to use.	Functional TBI with minor usability concerns.	Clear and effective TBI with structured user interaction.	Well-structured and user-friendly TBI, providing a seamless experience. Clearly



Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
					demonstrated in the report..
Sorting Implementation (4 marks)	Sorting algorithm is missing or incorrect.	Basic sorting algorithm implemented but not fully functional.	Sorting algorithm works but has inefficiencies.	Efficient sorting implemented with minor optimization issues.	Correct and well-optimized sorting algorithm used appropriately. Clearly demonstrated in the report..
Searching Implementation (4 marks)	Searching algorithm is missing or incorrect.	Basic searching algorithm implemented but not fully functional.	Searching algorithm works but has inefficiencies.	Efficient searching implemented with minor optimization issues.	Correct and well-optimized searching algorithm used appropriately. Clearly demonstrated in the report..
Error Handling (2 marks)	Exception handling is missing or incorrect.	Basic error handling is implemented but lacks robustness.	Good exception handling, but some edge cases are not covered.	Effective exception handling with minor gaps.	Comprehensive error handling with clear error messages. Clearly demonstrated in the report..
Test Cases (3 marks)	No testing evidence provided or test cases are incorrect.	Basic test cases provided with limited coverage.	Adequate test cases covering common scenarios.	Comprehensive testing with well-documented results.	Extensive testing covering most edge cases, with clear documentation and debugging logs. Clearly demonstrated in the report..
Code Comments and Readability (2 marks)	Code is unreadable, poorly organized, and lacks comments	Code is somewhat organized but lacks appropriate comments	Code is reasonably organized with basic comments	Code is well-organized with minor missing comments or structure improvements	Code is very well-organized, clear, and well-commented. Clearly demonstrated in the report..
Algorithm Complexity Discussion (3 marks)	No discussion on algorithm complexity or flaws	Basic discussion on algorithm complexity with limited understanding	Reasonable discussion on algorithm complexity.	Clear discussion on algorithm complexity	Thorough and insightful discussion on algorithm complexity
Report and Screenshots (5 marks)	Missing or poorly structured report and screenshots	Basic report and screenshots, but with significant issues	Reasonable report with basic screenshots; some issues remain. Shows a reasonable number of appropriate code snippets and outputs but missing figure	Well-organized report with adequate screenshots; minor gaps. Shows a reasonable number of appropriate code snippets and outputs with figure	Well-organized and thorough report with detailed screenshots. Shows most appropriate code snippets and outputs with figure captions and figure numbers.



Criteria	Fail (0 – 49%)	Pass (50 – 64%)	Credit (65 – 74%)	Distinction (75 – 84%)	High Distinction (85 – 100%)
			captions and figure numbers.	captions and figure numbers.	
Clarity of Demonstration (5 marks)	Demonstration is unclear, missing key points, or lacks structure or absent	Basic demonstration; some aspects unclear or missing	Clear demonstration but some aspects could be explained better	Mostly clear and well-explained demonstration with minor gaps	Excellent and clear demonstration of all aspects of the project
Total mark out of 35					

3.2 General information about assessment

a) Late Penalties and Extensions

An important part of business life and key to achieving KOI's graduate outcome of Professional Skills is the ability to manage workloads and meet deadlines. Completing assessment tasks on time is a good way to master these habits.

Students who miss mid-trimester tests and final exams without a valid and accepted reason may not be granted a deferred exam and will be awarded 0 marks for the assessment item. Assessment items which are missed or submitted after the due date/time will attract a penalty unless there is a compelling reason (see below). These penalties are designed to encourage students to develop good time management practices, and to create equity for all students.

Any penalties applied will only be up to the maximum marks available for the specific piece of assessment attracting the penalty.

Late penalties, granting of extensions and deferred exams are based on the following:

In Class Tests and Quizzes (excluding Mid-Trimester Tests)

- Generally, extensions are not permitted. A make-up test may only be permitted under very special circumstances where acceptable supporting evidence of illness, hardship or unavoidable problems preventing completion of the assessment is provided (see section (b) below). The procedures and timing to apply for a make-up test (only if available) are as shown in the section *Applying for an Extension* (see below).
- Missing a class test will result in 0 marks for that assessment item unless the above applies.

Written Assessments and Video Assessments

- There is a late penalty of 5% of the total available marks per calendar day unless an extension is approved (see *Applying for an Extension* section below).

Presentations

- Generally, extensions are not permitted. Missing a presentation will result in 0 marks for that assessment item. The rules for make-up presentations are the same as for missing in-class tests (described above).

For group presentations, if serious circumstances prevent some members of the group from participating, the members of the group who are present should make their contributions as agreed. If a make-up presentation is approved, the other members of the group will be able to make their individual presentation later and will be marked according to the marking rubric. A video presentation may be used to facilitate the process.



Mid-Trimester Tests and Final Exams

If students are unable to attend mid-trimester tests or final exams due to illness, hardship or some other unavoidable problem (acceptable to KOI), they must:

- Complete the Assignment Extension / Exam Deferment Form available by contacting academic@koi.edu.au as soon as possible, *but no later than three (3) working days after the exam date*.
- Provide acceptable documentary evidence (see section (b) below).
- Agree to attend the deferred exam as set by KOI if a deferred exam is approved.

Deferred exam

- There will only be one deferred exam offered.
- Marks obtained for the deferred exam will be the marks awarded for that assessment item.
- If you miss the deferred exam you will be awarded 0 marks for the assessment item. This may mean you are unable to pass the subject.

b) Applying for an Extension

If students are unable to submit or attend an assessment when due, they must

- Complete the Assignment Extension / Exam Deferment Form available by contacting academic@koi.edu.au as soon as possible, *but no later than three (3) working days of the assessment due date*.
- Provide acceptable documentary evidence in the form of a medical certificate, police report or some other appropriate evidence of illness or hardship, or a technician's report on problems with computer or communications technology, or a signed and witnessed statutory declaration explaining the circumstances.
- Students and lecturers / tutors will be advised of the outcome of the extension request as soon as practicable.

Please remember there is no guarantee of an extension being granted, and poor organisation is not a satisfactory reason to be granted an extension.

c) Referencing and Plagiarism

Please remember that all sources used in assessment tasks must be suitably referenced.

Failure to acknowledge sources is plagiarism, and as such is a very serious academic issue. Students plagiarising run the risk of severe penalties ranging from a reduction in marks through to 0 marks for a first offence for a single assessment task, to exclusion from KOI in the most serious repeat cases. Exclusion has serious visa implications. The easiest way to avoid plagiarising is to reference all sources.

Harvard referencing is the required method – in-text referencing using Author's Surname (family name) and year of publication. A Referencing Guide, "Harvard Referencing", and a Referencing Tutorial can be found on the right-hand menu strip in Moodle on all subject pages.

An effective way to reference correctly is to use *Microsoft Word's* referencing function (please note that other versions and programs are likely to be different). To use the referencing function, click on the References Tab in the menu ribbon – students should choose *Harvard*.

Authorship is also an issue under plagiarism – KOI expects students to submit their own original work in both assessment and exams, or the original work of their group in the case of a group project. All students agree to a statement of authorship when submitting assessments online via Moodle, stating that the work submitted is their own original work.

The following are examples of academic misconduct and can attract severe penalties:



- Handing in work created by someone else (without acknowledgement), whether copied from another student, written by someone else, or from any published or electronic source, is fraud, and falls under the general Plagiarism guidelines.
- Copying / cheating in tests and exams is academic misconduct. Such incidents will be treated just as seriously as other forms of plagiarism.
- Students who willingly allow another student to copy their work in any assessment may be considered to assisting in copying/cheating, and similar penalties may be applied.

Where a subject coordinator considers that a student might have engaged in academic misconduct, KOI may require the student to undertake an additional oral exam as a part of the assessment for the subject, as a way of testing the student's understanding of their work.

Further information can be found on the KOI website.

d) Reasonable Adjustment

The Commonwealth Disability Discrimination Act (1992) makes it unlawful to treat people with a disability less fairly than people without a disability. In the context of this subject, the principle of Reasonable Adjustment is applied to ensure that participants with a disability have equitable access to all aspects of the learning for the subject. For assessment, this means that barriers to their demonstrating competence are removed wherever it is reasonably practical to do so.

Examples of reasonable adjustment in assessment may include:

- provision of an oral assessment, rather than a written assessment
- provision of extra time
- use of adaptive technology.

The focus of the adjusted assessment should be on enabling the student to demonstrate achievement of the learning outcomes for the subject, rather than on the method of assessment.

e) Appeals Process

Full details of the KOI *Assessment and Assessment Appeals Policy* may be obtained in hard copy from the Library, and on the KOI website www.koi.edu.au under *Policies and Forms*.

Assessments and Mid-Trimester Exams:

Where students are not satisfied with the results of an assessment, including mid-trimester exams, they have the right to appeal. The process is as follows:

- Discuss the assessment with their tutor or lecturer – students should identify where they feel more marks should have been awarded – students should provide valid reasons based on the marking guide provided for the assessment. Reasons such as “*I worked really hard*” are not considered valid.
- If still not satisfied, students should complete an Application for Review of Assessment Marks form, clearly explaining the reasons for seeking a review. This form is available from the KOI website under *Policies and Forms* and is also available at KOI Reception (Kent St, Market St and O’Connell St). The completed Application for Review of Assessment Marks form should be submitted as explained on the form with supporting evidence attached to academic@koi.edu.au.
- The form must be submitted within *ten (10) working days* of the return of the marked assessment, or within *five (5) working days* after the return of the assessment if the assessment is returned after the end of the trimester.

Review of Grade – whole of subject and final exams:

Where students are not satisfied with the results of the whole subject or with their final exam results, they have the right to request a Review of Grade – see the *Assessment and Assessment Appeals Policy* for more information.



An *Application for Review of Grade/Assessment Form* (available from the KOI Website under *Policies and Forms* and from KOI Reception at Kent St, Market St and O'Connell St) should be completed clearly explaining the grounds for the application. The completed application should be submitted as explained on the form, with supporting evidence attached to academic@koi.edu.au